

Multiple Choice

1. **Answer:** A

Options: A) Levels of oxygen, acidity or alkalinity of the medium; B) Temperature changes, light exposure; C) Atmospheric pressure, noise levels; D) Nitrogen levels, radiation exposure; E) Soil texture, light wavelength

Note: Correct option: A - Levels of oxygen, acidity or alkalinity of the medium

2. **Answer:** C

Options: A) Lysosomes act as storage units for excess nutrients; B) Lysosomes are used for energy production; C) Lysosomes function in intra cellular digestion; D) Lysosomes synthesize proteins for the cell; E) Lysosomes regulate cell division

Note: Correct option: C - Lysosomes function in intra cellular digestion

3. **Answer:** A

Options: A) Only two DNA regions, V and D, combine in heavy chains; B) V, D, and JH regions are combined in light chains; C) V and JH regions are combined, while D regions are excluded from heavy chains; D) Five separate DNA regions are combined in heavy chains, adding extra complexity; E) D and JH regions are combined, while V regions are excluded from heavy chains

Note: Correct option: A - Only two DNA regions, V and D, combine in heavy chains

4. **Answer:** A

Options: A) 12.5 percent; B) 40 percent; C) 18.75 percent; D) 25 percent; E) 37.5 percent

Note: Correct option: A - 12.5 percent

5. **Answer:** E

Options: A) - .09176; B) .21185; C) .37748; D) .55962; E) .65361

Note: Correct option: E - .65361

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'directional selection'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'inheritance of both "fit" and "unfit" genes'.

8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Innate behavior'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'clathrin'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The size of the appendix varies greatly among individuals.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** A flock of birds can be considered an elementary form of society called a motion group.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key